BTS-VI(R/S)-06-23-1815 Reg. No. | | | | | | | | |

A

B.Tech. Degree VI Semester Regular/Supplementary Examination June 2023

ME 19-205-0604 HEAT AND MASS TRANSFER

(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the basic modes of heat transfer.
CO2: Get an insight on conduction, convection and radiation heat transfers.
CO3: Apply the knowledge of heat transfer through various geometries for optimizing the thickness of insulation.
CO4: Understand the unsteady heat conduction and its applications.
CO5: Get a concept on multiphase flow, diffusion and convective mass transfer.
CO6: Attain information of parallel and counterflow heat exchangers and their design aspects.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	State and explain Fourier's law of heat conduction.	3	L2	1	1
	(b)	Define Biot number. What is its significance?	3	L1	4	1
	(c)	How we can differentiate thermal and hydrodynamic boundary layers?	3	L2	5	1
	(d)	Why are higher heat transfer rates experienced in drop wise condensation than in film condensation?	3	L2	5	1
	(e)	State and explain Kirchhoff's law.	3	L1	2	1
	(f)	Explain the terms radiosity and irradiation with a simple sketch.	3	L1	2	1
	(g)	What do you mean by fouling factor? Mention the causes of fouling.	3	L2	6	1
	(h)	Compare the parallel flow and counterflow heat exchanger. Why the counterflow heat exchangers are commonly used?	3	L2	6	1

PART B

$(4 \times 12 = 48)$

			Marks	BL	CO	PO
II.	(a)	Derive two-dimensional heat conduction equation in Cartesian coordinates. State the assumptions used. Represent the same in vector form.	6	L3	1,2	2
	(b)	A composite wall of a house is made of 0.1 m layer of common brick (k = 0.7 W/m°C) followed by a 0.04 m layer of gypsum plaster (k = 0.48 W/m°C). What thickness of loosely packed rock wool insulation (k = 0.065 W/m°C) should be added to reduce the heat loss or gain through the wall by 80%?	6	L3	1,2	2
		OR				
III.	(a)	State and explain the critical thickness of insulation based on electrical wires and a steam pipe. Derive an expression for critical thickness of insulation for a cylinder.	6	L3	3	2
	(b)	A wire of 7 mm diameter at a temperature of 65°C is to be insulated by a material having thermal conductivity k = 0.174 W/m°C. The convective heat transfer coefficient h_o = 8.712 W/m^2 °C. The ambient temperature is 20°C. Find the critical radius of insulation. For maximum heat loss, what is the minimum thickness of insulation and the heat loss per metre length? Also compare the amount of heat dissipated with this thickness of insulation and without insulation.	6	L3	3	3

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			Marks	BL	CO	PO
IV.	(a)	Explain the boiling regimes with a suitable sketch.	6	L2	5	2
	(b)	Water at the rate of 0.6 kg/s is passed through a tube of 20 mm diameter. Then it is found to be heated from 20°C to 50°C. The heating is accomplished by condensing steam on the surface of the tube and subsequently the surface temperature of the tube is maintained at 85°C. Determine the length of the tube required for fully developed flow. Take the thermos physical properties of water at 60°C as: $\rho = 983.2$ kg/m^3; $C_p = 4.178$ kJ/kgK, $k = 0{,}659$ W/m°C; kinematic viscosity $\nu = 0.478 \times 10^{-6}$ m^2/s.	6	L3	5	2
		OR				
V.	(a)	Define Reynolds Analogy.	5	L1	5	1
	(b)	A cylindrical body of 300 mm diameter and 1.6 m height is maintained at a constant temperature of 36.5°C. The surrounding temperature is 13.5°C. Find out the amount of heat to be generated by the body per hour if: $\rho = 1.025$ kg/m^3, Cp = 0.96 KJ/kg°C, $\nu = 15.06 \times 10^{-6}$ m^2/s, $k = 0.0892$ kJ/m-sec°C, and $\beta = 0.003356$ K^{-1}. Assume $Nu = 0.12\,(Gr.\,Pr)^{1/3}$	7	L3	5	2
VI.	(a)	Write a note on the properties of black body. What potential changes in heat transfer occur if a surface selected is gray instead of black surface?	5	L2	2	1
	(b)	Assuming the sun to be a black body having a surface temperature of 5800 K, calculate: (i) the total emissive power (ii) the wavelength at which the maximum spectral intensity occurs (iii) the maximum value of $E_{b\lambda}$.(iv) the total energy emitted by the sun per unit time if its diameter can be assumed to be 1.391×10^{9} m.	7	L3	2	2
		OR				
VII.	(a)	Explain the effect of using radiation shield. What is the criteria for selecting shield materials?	4	L2	2	1
	(b)	Two large plates are placed parallel, one at temperature t_1 = 827°C with emissivity $\varepsilon_1 = 0.8$ and other at t_2=327°C with emissivity $\varepsilon_2 = 0.4$. An aluminium radiation shield with an emissivity $\varepsilon_s = 0.05$ on both sides is placed between the plates. Calculate the percentage reduction in heat transfer rate between the two plates as a result of the shield.	8	L3	2	2
VIII.	(a)	Derive the expression for LMTD for a counter flow heat exchanger.	6	L2	6	2
	(b)	The flow rates of hot and cold water streams running through a parallel flow heat exchanger are 0.2 kg/s and 0.5 kg/s respectively. The inlet temperatures on the hot and cold sides are 75°C and 20°C respectively. The exit temperature of hot water is 40°C. If the individual heat transfer coefficients on both sides are 650 W/m^2°C, calculate the area of the heat exchanger.	6	L3	6	3
		OR				
IX.	(a)	How are the heat exchangers classified?	2	L1	6	1
	(b)	In an oil cooler oil enters at 170°C. If water entering at 30°C flows parallel to the oil, the exit temperature of oil and water are 90°C and 70°C respectively. Determine the exit temperature of oil and water if the two fluids flow in the opposite directions. Assume that the flow rates of the two fluids and U_O remain unaltered. What would be the minimum temperatures to which the oil could be cooled in parallel flow operation?	10	L3	6	2

Bloom's Taxonomy Levels
L1 = 20.83%, L2 = 37.5%, L3 = 41.67%.

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